

Capstone Project Practice

Hands-on application of Python to solve real business problems through an end-to-end AI & Data Analytics capstone project.



Project Introduction

Retail businesses generate large volumes of transactional data every day. Analyzing this data helps companies understand **customer purchasing patterns, product performance, seasonal trends, and regional sales behavior**. These insights support better decision-making in **inventory management, marketing strategy, pricing, and profit optimization**.

In this project, we perform an **end-to-end sales data analysis** using Python. The goal is to clean, explore, and visualize retail sales data to answer important business questions such as:

- Which product categories generate the most revenue?
- How do sales vary over time (monthly trends)?
- Which regions contribute most to overall sales?
- How do discounts impact profitability?
- What does the distribution of sales values look like?



Structured data analytics pipeline:

This project follows a structured data analytics pipeline:

1. **Data Acquisition** – Loading a real-world retail dataset
2. **Data Preparation** – Cleaning and transforming the data
3. **Data Analysis** – Calculating business metrics and patterns
4. **Visualization** – Creating charts to tell a clear business story

The project uses beginner-friendly Python libraries:

- **Pandas** for data handling
- **Matplotlib & Seaborn** for visualizations



Dataset Information

Dataset Name

Superstore Sales Dataset (commonly available on Kaggle and GitHub)

Dataset Description

This dataset contains historical sales records of a retail superstore. Each row represents an individual order transaction including product details, customer region, pricing, and profit.

It is widely used for beginner data analytics projects because it is:

- Realistic but easy to understand
- Clean and well-structured
- Suitable for trend, comparison, and distribution analysis



Step 1:

```
# =====  
# Retail Sales Data Analysis (Beginner Project)  
# Tools: pandas, matplotlib, seaborn  
# Stages: Acquisition -> Preparation -> Analysis -> Visualization  
# =====  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
sns.set(style="whitegrid") # consistent styling for all charts  
  
# -----  
# Stage 1: Data Acquisition  
# -----  
df = pd.read_csv('Dataset/superstore_sample_dataset.csv', encoding="ISO-8859-1")  
df.columns = df.columns.str.strip() # safe  
print ("----- Column Name----- " )  
print(df.columns)  
  
print("Rows, Columns:", df.shape)  
print("\n--- First 5 rows ---")  
print(df.head())
```

Step 2

```
# -----  
# Step 2: Basic Cleaning  
# -----  
# Convert dates to datetime  
df["Order Date"] = pd.to_datetime(df["Order Date"])  
df["Ship Date"] = pd.to_datetime(df["Ship Date"])  
  
# Remove duplicates (if any)  
df = df.drop_duplicates()  
  
# If any important values are missing, drop those rows (basic)  
df = df.dropna(subset=["Order Date", "Sales"])  
  
# Create Month-Year for trend analysis  
df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)
```

Step 3:

```
# -----  
# Step 3: Simple Analysis Tables  
# -----  
print("\nTotal Sales:", df["Sales"].sum())  
print("Total Profit:", df["Profit"].sum())  
print("Average Sales:", df["Sales"].mean())  
  
monthly_sales = df.groupby("YearMonth")["Sales"].sum().reset_index()  
category_sales = df.groupby("Category")["Sales"].sum().reset_index()  
region_sales = df.groupby("Region")["Sales"].sum().reset_index()  
  
top_subcat = (  
    df.groupby("Sub-Category")["Sales"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
        .reset_index()  
)
```

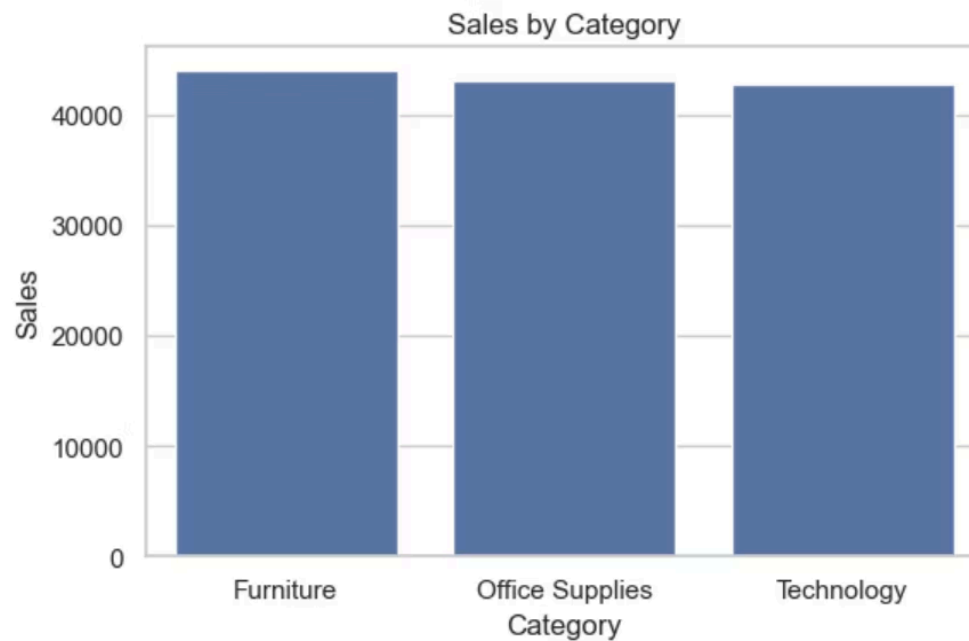
Step 4: visualization

```
# 1) Monthly Sales Trend
plt.figure(figsize=(10,4))
plt.plot(monthly_sales["YearMonth"], monthly_sales["Sales"], marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



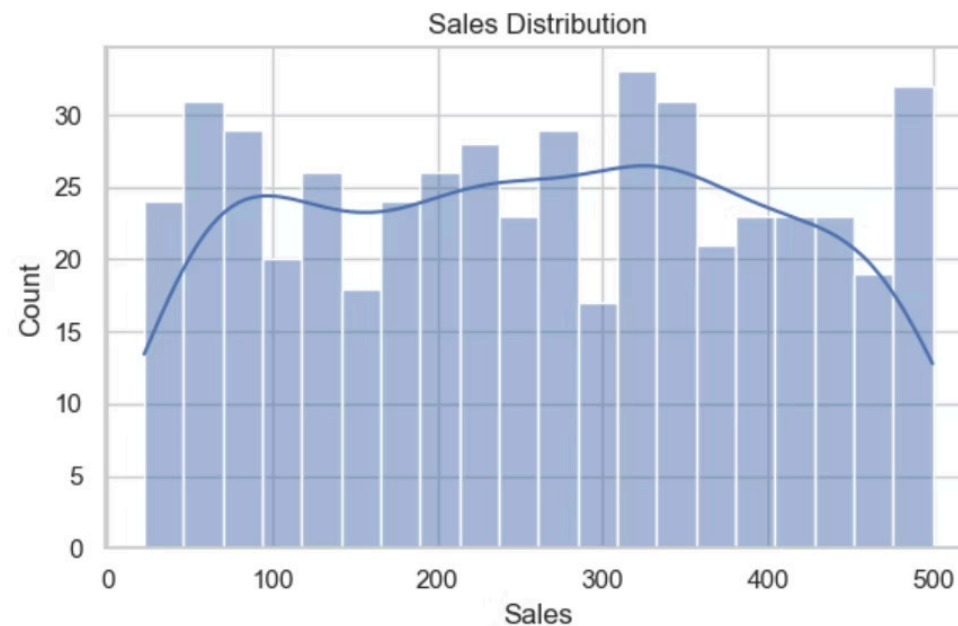
Step 4: visualization

```
# 2) Sales by Category
plt.figure(figsize=(6,4))
sns.barplot(data=category_sales, x="Category", y="Sales")
plt.title("Sales by Category")
plt.tight_layout()
plt.show()
```



Step 4: visualization

```
# 3) Sales Distribution  
plt.figure(figsize=(6,4))  
sns.histplot(df["Sales"], bins=20, kde=True)  
plt.title("Sales Distribution")  
plt.tight_layout()  
plt.show()
```



Execute the code in Jupyter Notebook

Case Study 1

Marketing Campaign Performance Analysis

Case Study 2



Business Problem

A company ran multiple marketing campaigns (Email, Social Media, Google Ads, etc.).

They want to understand:

- Which campaign type brings the most conversions?
- Which age group responds the most?
- Does higher spending always lead to better results?
- What is the overall campaign success rate?



Dataset Information

Column	Meaning
Customer ID	Unique customer number
Age	Age of customer
Gender	Male/Female
Campaign Type	Email, Social Media, Google Ads, TV
Ad Spend	Amount spent on showing ads
Clicks	Number of times ad was clicked
Conversions	Whether the customer purchased (0 = No, 1 = Yes)
Date	Date of interaction

Step 1:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# Stage 1: Data Acquisition
# -----
df = pd.read_csv('Dataset/marketing_campaign.csv', encoding="ISO-8859-1")
df.columns = df.columns.str.strip() # safe
print("----- Column Name----- ")
print(df.columns)

print("Rows, Columns:", df.shape)
print("\n--- First 5 rows ---")
print(df.head())
```

```
----- Column Name-----
Index(['Customer ID', 'Age', 'Gender', 'Campaign Type', 'Ad Spend', 'Clicks',
      'Conversions', 'Date'],
      dtype='object')
Rows, Columns: (400, 8)
```

```
--- First 5 rows ---
Customer ID  Age  Gender Campaign Type  Ad Spend  Clicks  Conversions \
0  CUST0099   57  Female    Google Ads    88.97     8         0
1  CUST0231   52   Male    Google Ads    81.93     9         0
2  CUST0018   30  Female  Social Media    59.01     8         0
3  CUST0084   60  Female         Email    74.31    19         1
4  CUST0107   44   Male         Email    51.46    17         0

Date
0  2023-06-26
1  2023-05-04
2  2023-02-26
3  2023-09-06
4  2023-12-26
```

Step 2:

```
# Step 2: EDA - Data Preparation

# Convert date column
df["Date"] = pd.to_datetime(df["Date"])

# Remove duplicates
df = df.drop_duplicates()

# Drop rows with missing important values
df = df.dropna(subset=["Campaign Type", "Ad Spend", "Conversions"])

# Create Age Group column
df["Age Group"] = pd.cut(df["Age"],
                          bins=[18,25,35,50,65],
                          labels=["18-25", "26-35", "36-50", "51-65"])
```

Step 3:

```
# Step 3: Analysis
```

```
# Key Metrics
```

```
total_spend = df["Ad Spend"].sum()  
total_conversions = df["Conversions"].sum()  
conversion_rate = (total_conversions / len(df)) * 100
```

```
print("Total Ad Spend:", total_spend)  
print("Total Conversions:", total_conversions)  
print("Conversion Rate (%)ate)", conversion_rate)
```

```
Total Ad Spend: 20982.92  
Total Conversions: 119  
Conversion Rate (%): 29.75
```

```
# Campaign Performance
```

```
campaign_performance = df.groupby("Campaign Type")["Conversions"].sum().reset_index()  
  
age_group_performance = df.groupby("Age Group", observed=True)["Conversions"].sum().reset_index()  
  
spend_by_campaign = df.groupby("Campaign Type")["Ad Spend"].sum().reset_index()
```

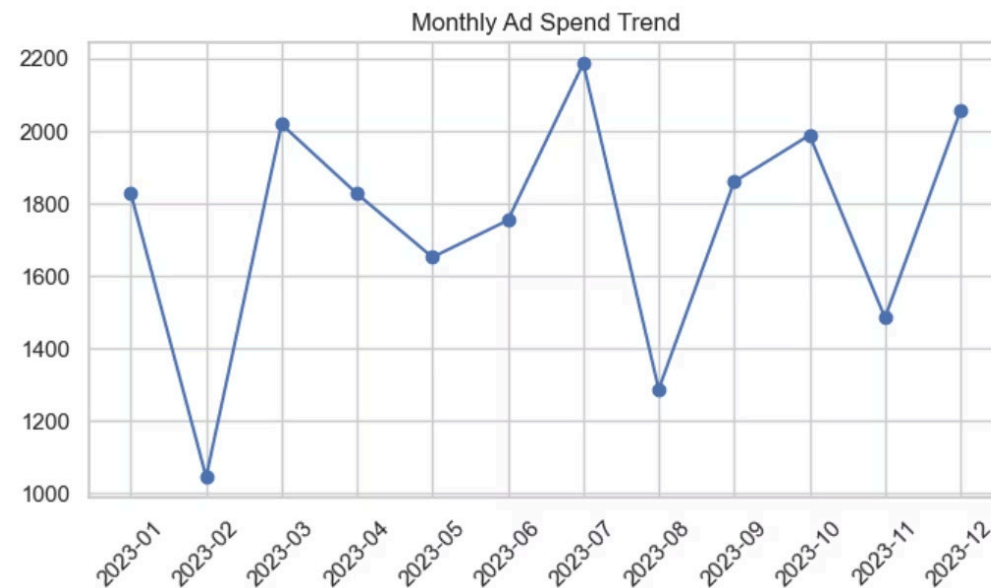
Step 4: Visualization

```
# Stage 4: Visualization

sns.set(style="whitegrid")

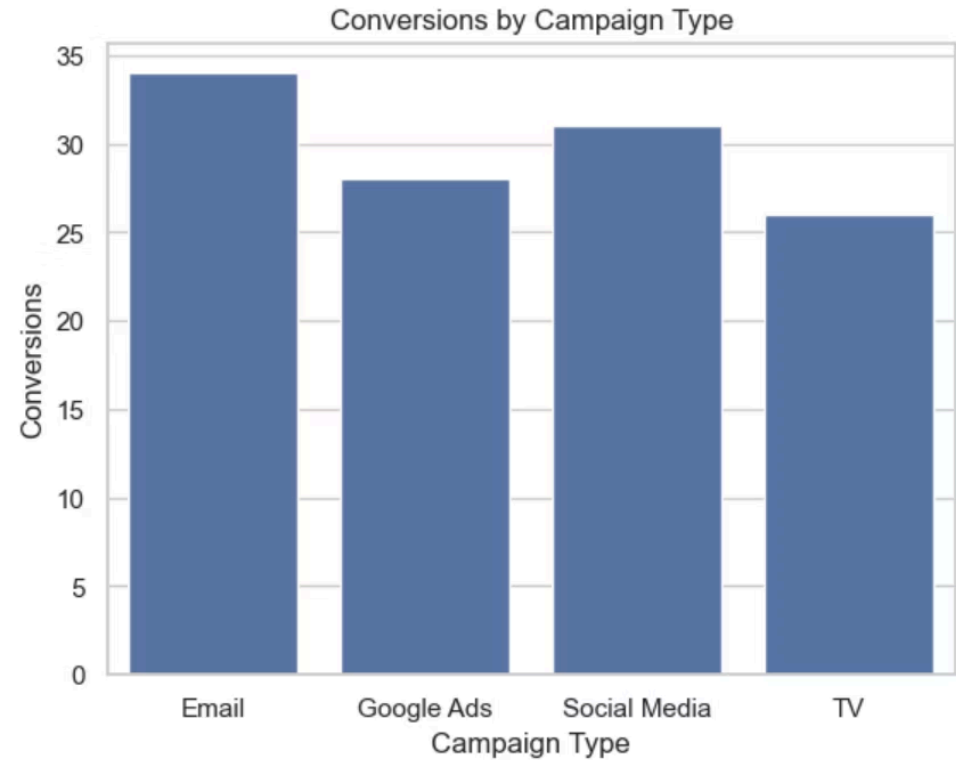
# Trend Chart
monthly_spend = df.groupby(df["Date"].dt.to_period("M"))["Ad Spend"].sum().reset_index()
monthly_spend["Date"] = monthly_spend["Date"].astype(str)

plt.figure(figsize=(8,4))
plt.plot(monthly_spend["Date"], monthly_spend["Ad Spend"], marker="o")
plt.title("Monthly Ad Spend Trend")
plt.xticks(rotation=45)
plt.show()
```



Step 4: Visualization

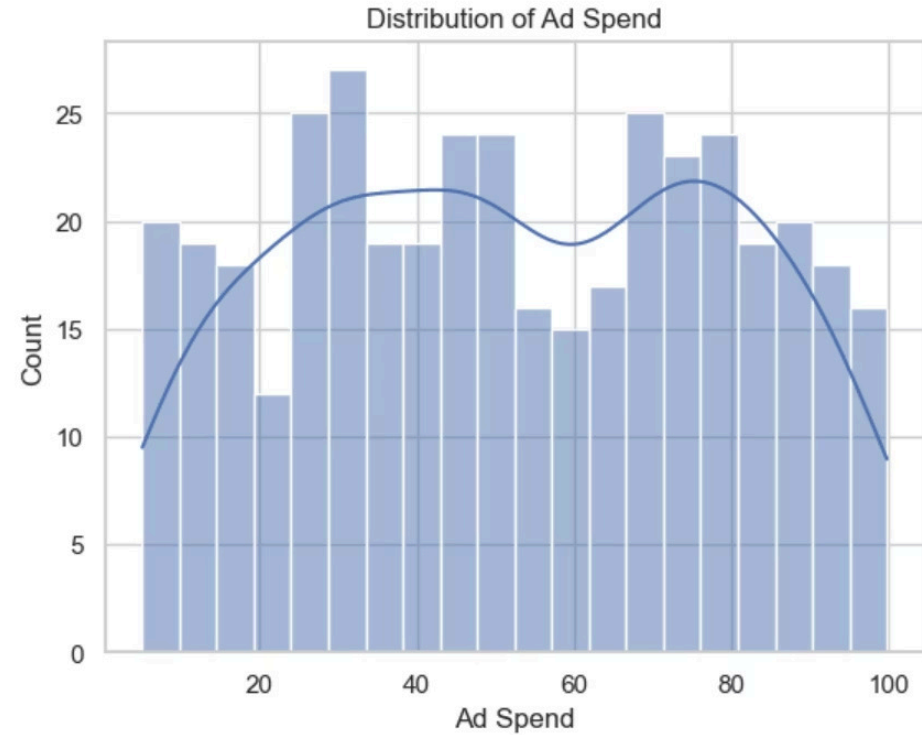
```
# Comparison Chart - Conversions by Campaign Type  
  
sns.barplot(data=campaign_performance, x="Campaign Type", y="Conversions")  
plt.title("Conversions by Campaign Type")  
plt.show()
```



Step 4: Visualization

```
# Distribution Chart - Ad Spend Distribution
```

```
sns.histplot(df["Ad Spend"], bins=20, kde=True)  
plt.title("Distribution of Ad Spend")  
plt.show()
```



Week 5: Final Project Dataset Options (Choose ONE)

Case Study Option	Domain	Business Problem to Solve	External Data Source	Key Analysis Focus	Required Visualizations (Minimum 3)	Simple Automation Idea
Retail Sales (Superstore)	Business Operations	Understand sales trends, top products, and regional performance to improve revenue decisions	Kaggle/GitHub Superstore dataset or provided sample CSV	Monthly sales trends, category performance, regional sales, profit vs discount	Trend: Monthly sales line chart, Comparison: Category sales bar chart, Distribution: Sales histogram+ Region sales chart+ Top products chart	Function to auto-generate monthly sales summary
Marketing Campaign Analysis	Marketing Analytics	Identify which campaigns generate the most conversions and which customer groups respond best	Provided marketing CSV or Kaggle campaign dataset	Conversion rate, campaign effectiveness, age group response, spend vs conversion	Trend: Monthly ad spend, Comparison: Conversions by campaign, Distribution: Ad spend histogram+ Age group chart+ Spend vs conversions chart	Auto-calculate weekly campaign performance report
Car Market Analysis (Car Dekho)	Business / Financial Analytics	Analyze factors affecting car selling prices and market demand	Provided Car Dekho dataset or Kaggle car price dataset	Price trends by year, fuel type demand, transmission preference, owner impact	Trend: Price vs year line chart, Comparison: Fuel type bar chart, Distribution: Price histogram+ Transmission chart+ Owner vs price chart	Function to summarize average car prices by year
Student Choice Dataset	Any Approved Domain	Student defines their own business question	Kaggle / GitHub / Public API	Must include grouping, metrics, and meaningful insights	Must include: Trend chart, Comparison chart, Distribution chart+ 2 relevant charts	Any automated summary or report generation